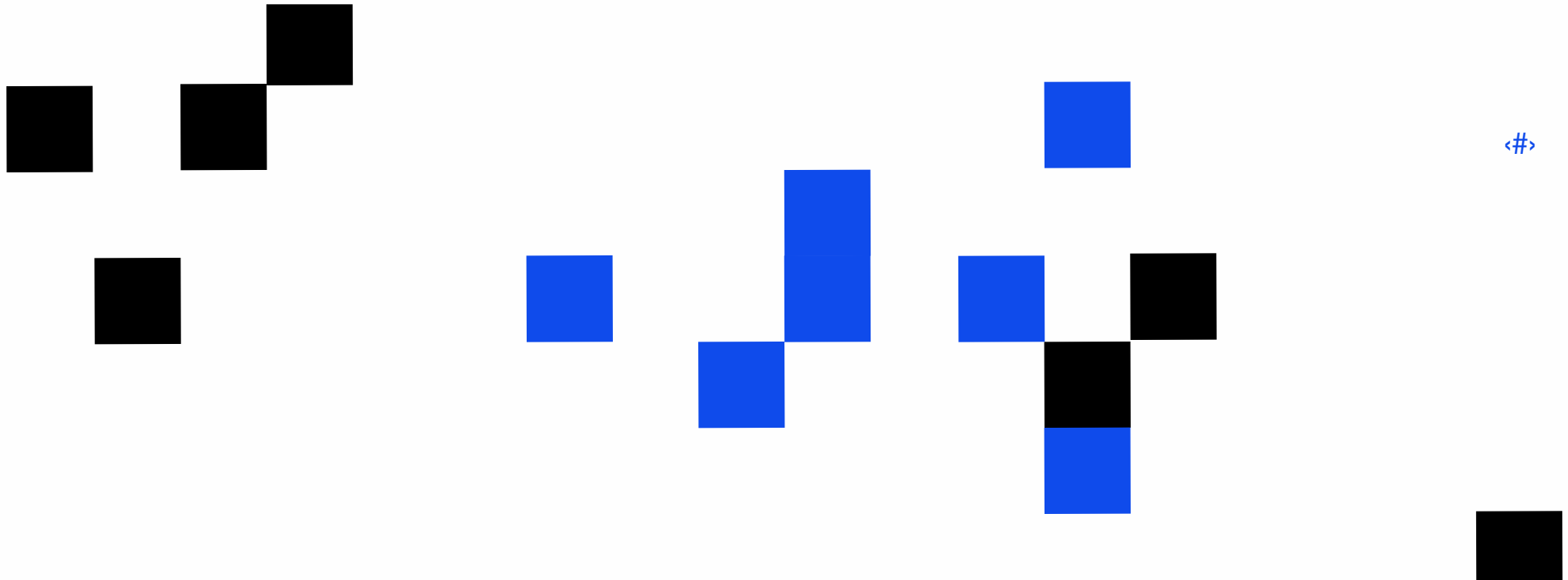


2026 Spring Software Innovation Studio

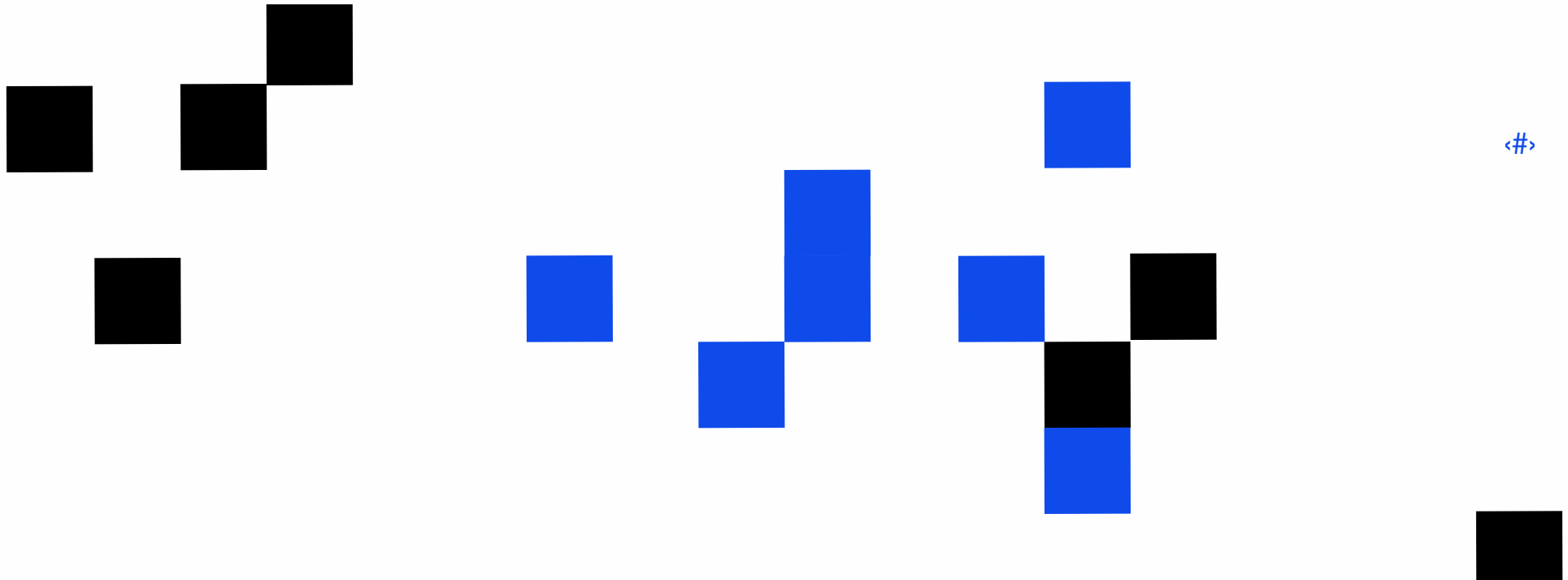
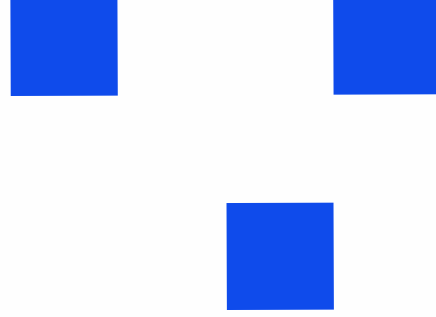
Dr. Yu-Kai Wang

Graphics created by
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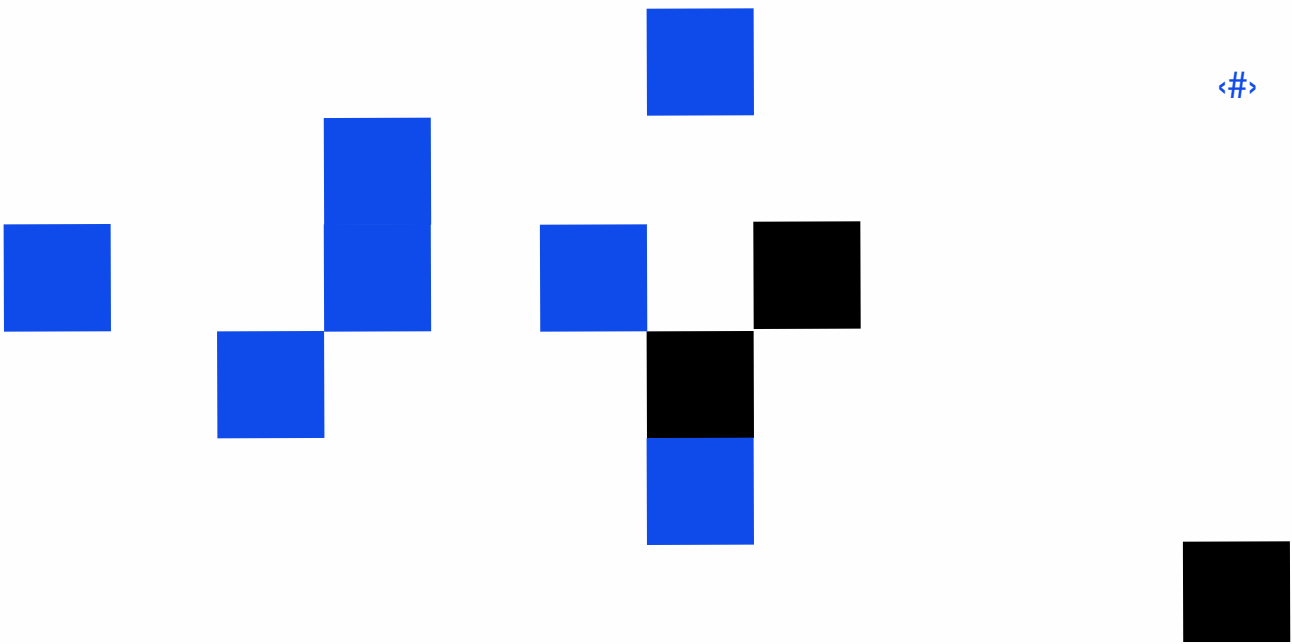
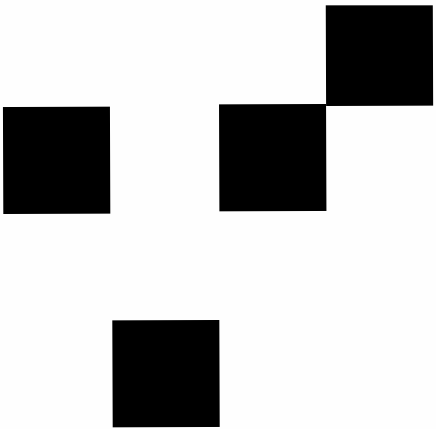
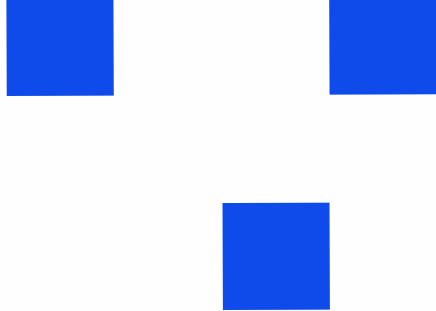
Agenda

- • **Intro the subject and high-level overview of assessment**
Teaching team introduction
Software Innovation Studio
- **Icebreaking and form the team**
- **Wrap the class**
Assessments overview
Homework

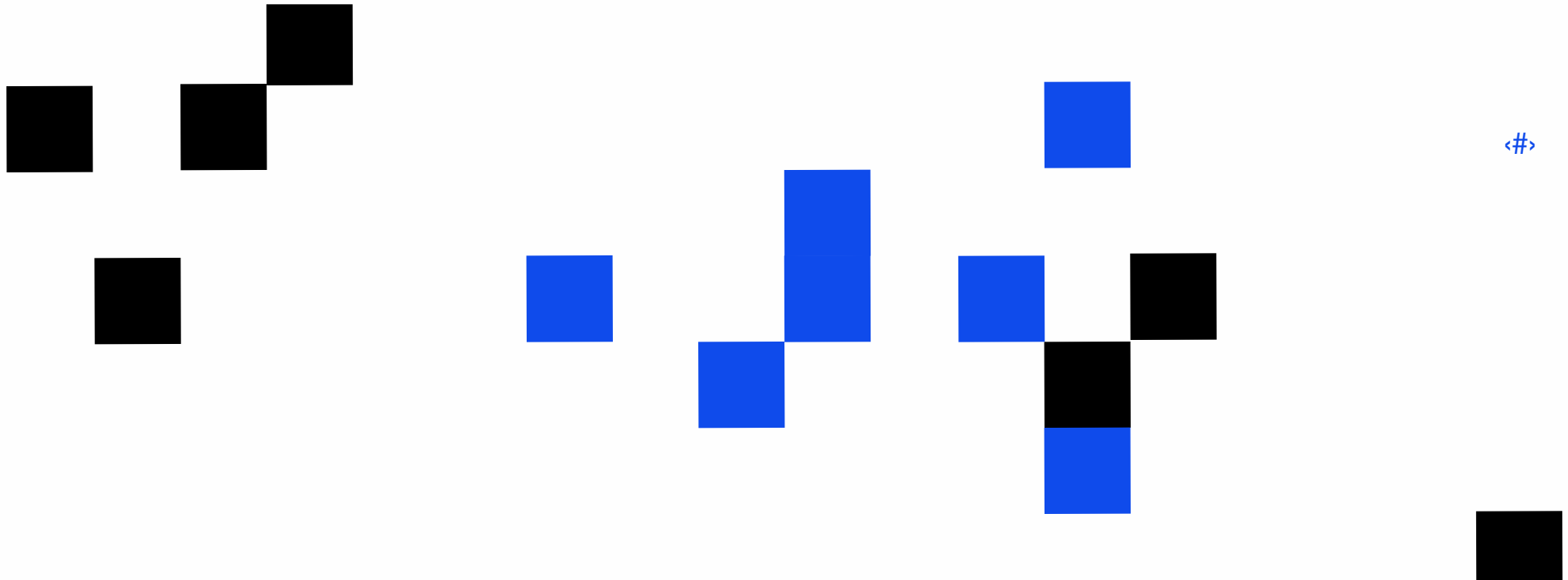
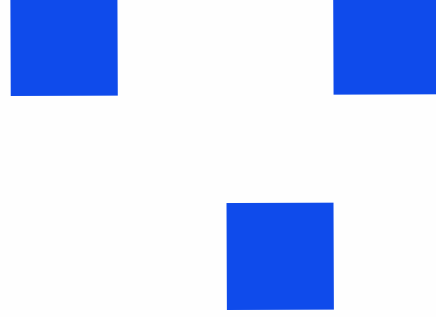


Session

- • **Workshop:**
 1. Campus (CB02.07.150)
 2. Friday 11:00 – 2 pm
- **Consultation / Q&A session:**
 1. In-person in classroom or my office, Friday 2 – 3 pm
 2. Feel free to bring questions and have discussions with teaching staff



Teaching Team



Teaching Team

- **Subject Coordinator:**

Dr. Yu-Kai (YK) Wang

■ School of Computer Science, FEIT

Director of Human-centric Augmentation and Intelligence Lab (HAIL)

Co-director of Motional Platform and Mixed Reality Lab (MPMR Lab)

Core Member of Australia Artificial Intelligence Institute

YuKai.Wang@uts.edu.au

- **Tutor:**

Connor Barkley, Zack Bailey, Ziyi Zhao, Tanzeem Syeda, and Cyrus Kwan

Connor.Barkley@uts.edu.au

Zack.Bailey@uts.edu.au

Ziyi.Zhao-1@uts.edu.au

ZahedaTanzeem.Syeda@uts.edu.au

Cyrus.Kwan@uts.edu.au

About Coordinator (YK Wang)

- Senior Lecturer in School of Computer Science, FEIT, UTS
- Core member in Australia Artificial Intelligence Institute (largest AI Institute in Australia, **global Top 10** for AI research)
- Human-centric Augmentation and Intelligence Lab (HAIL)
- Computation Intelligence and Brain Computer Interface (CIBCI) Lab
- Motion Platform and Mixed Reality (MPMR) Lab
- 10+ years in **AI**, **BCI**, **AR/VR** and **Human-Agent Teaming**



Immersive
Scenario

Bio
Sensing

Data
Analytics



c.



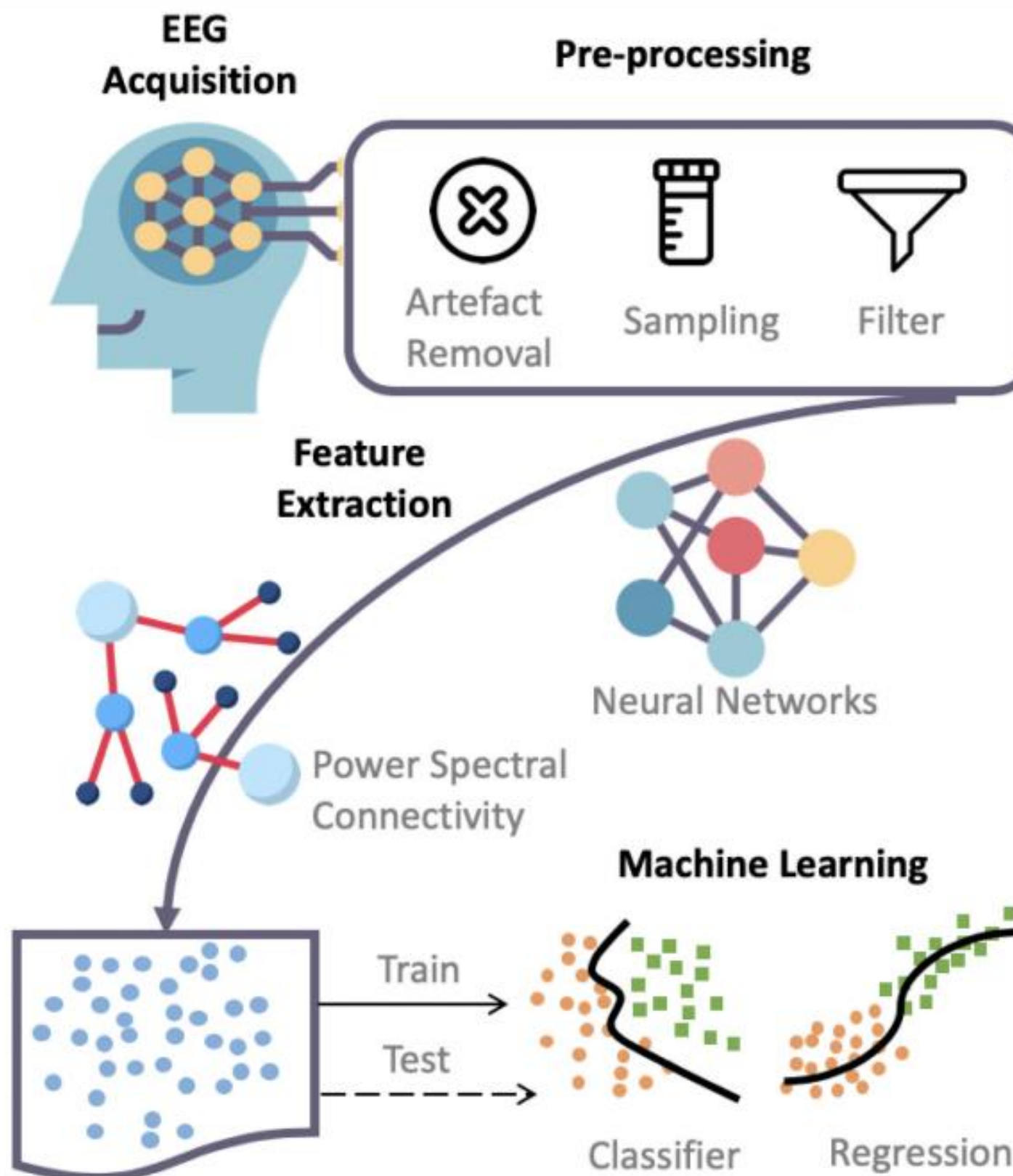
d.



h.



i.



Brain-Computer
Interface

AI modelling

Healthcare

Robot Control

About Tutor (Connor)

■ Graduate Biomedical Engineering (Hons)/Diploma of Professional Engineering Practice

- Capstone - Exploratory fatigue-based study of Human State Modelling to Increase the safety and reliability of Human-Autonomous Teaming

PhD Student, 3rd year - Language Independent Text Classification from EEG

- Intersection of linguistics, neuroscience, and machine learning
- Focusing on codifying and classifying information from the Semantic Language Network

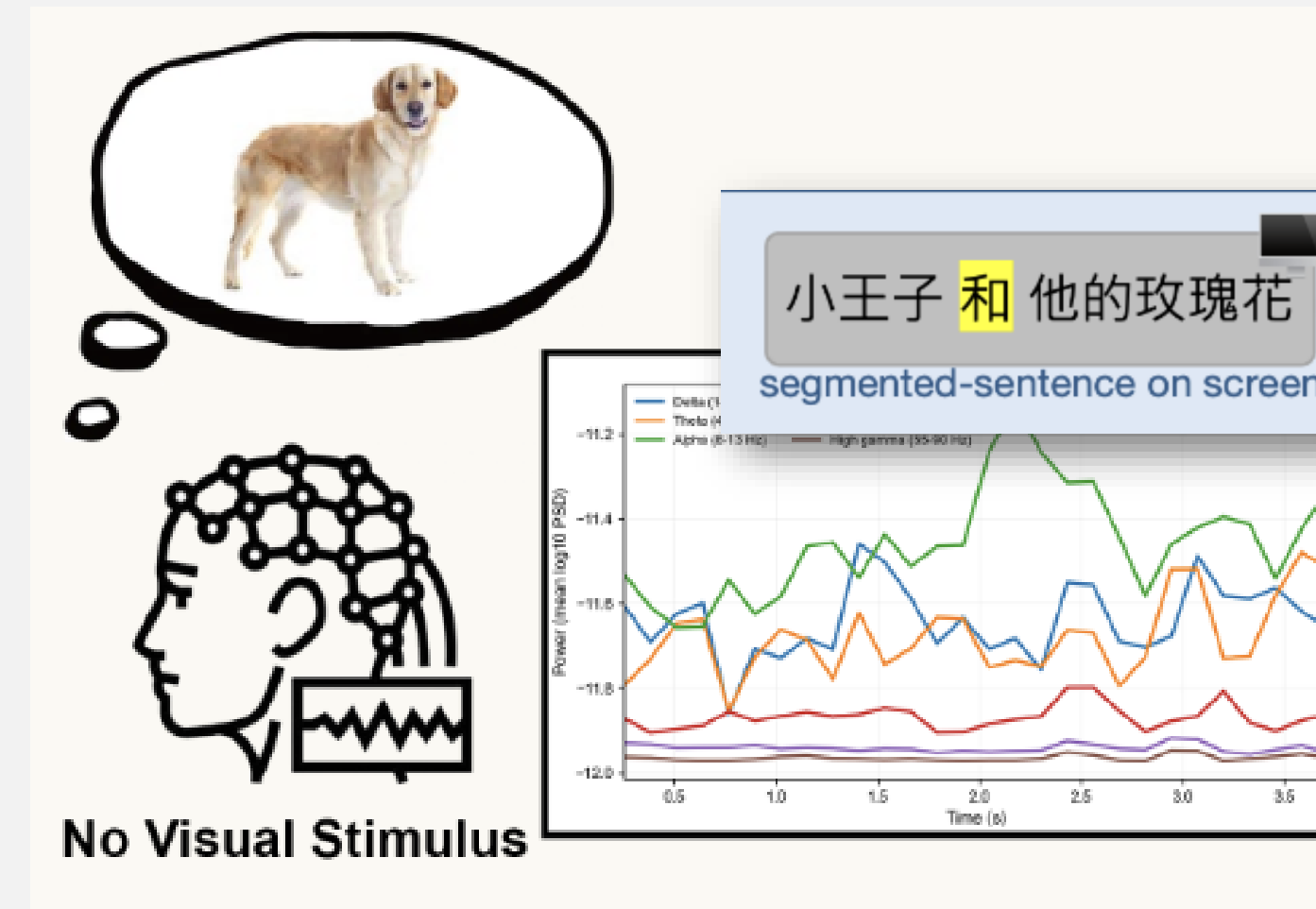
About Tutor (Ziyi)



PhD Candidate

Doing research in BCI

- ❖ EEG ---> Language
- ❖ EEG ---> Image
- ❖ EEG ---> Motion



Education



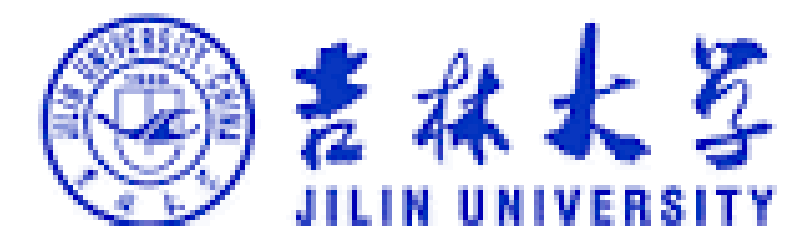
Switzerland



China



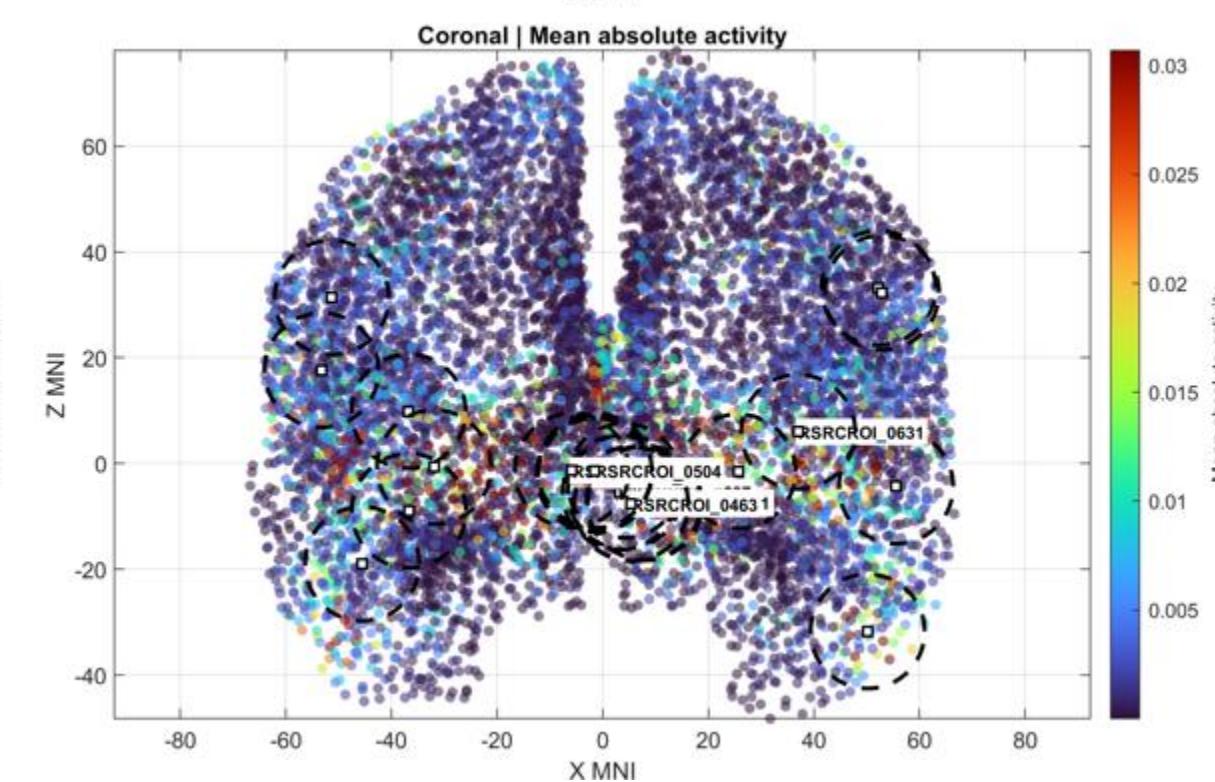
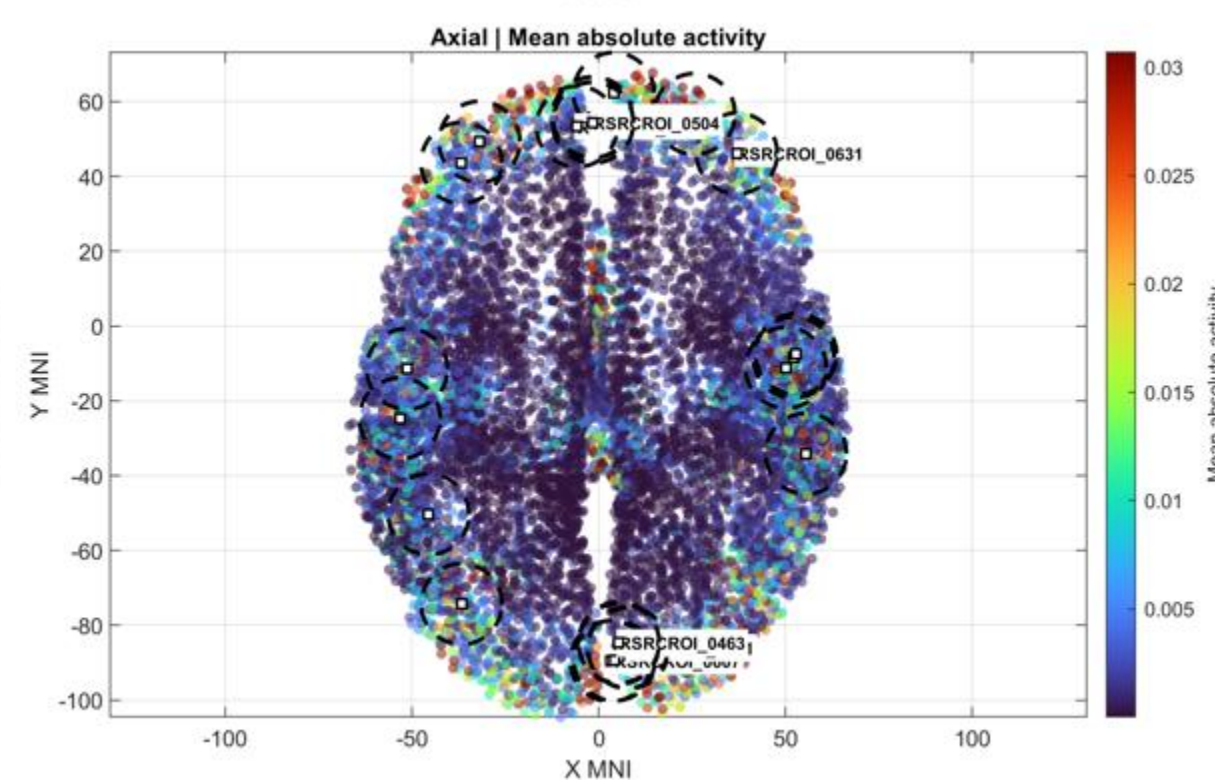
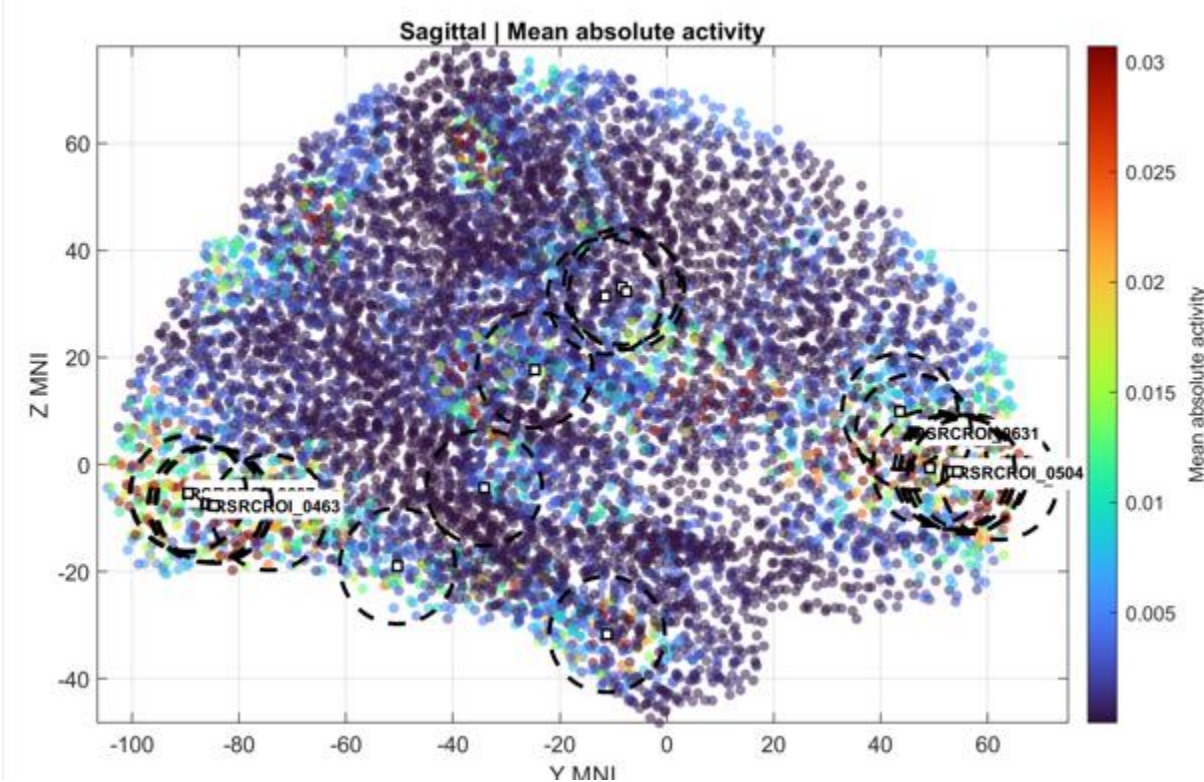
同济大学
TONGJI UNIVERSITY



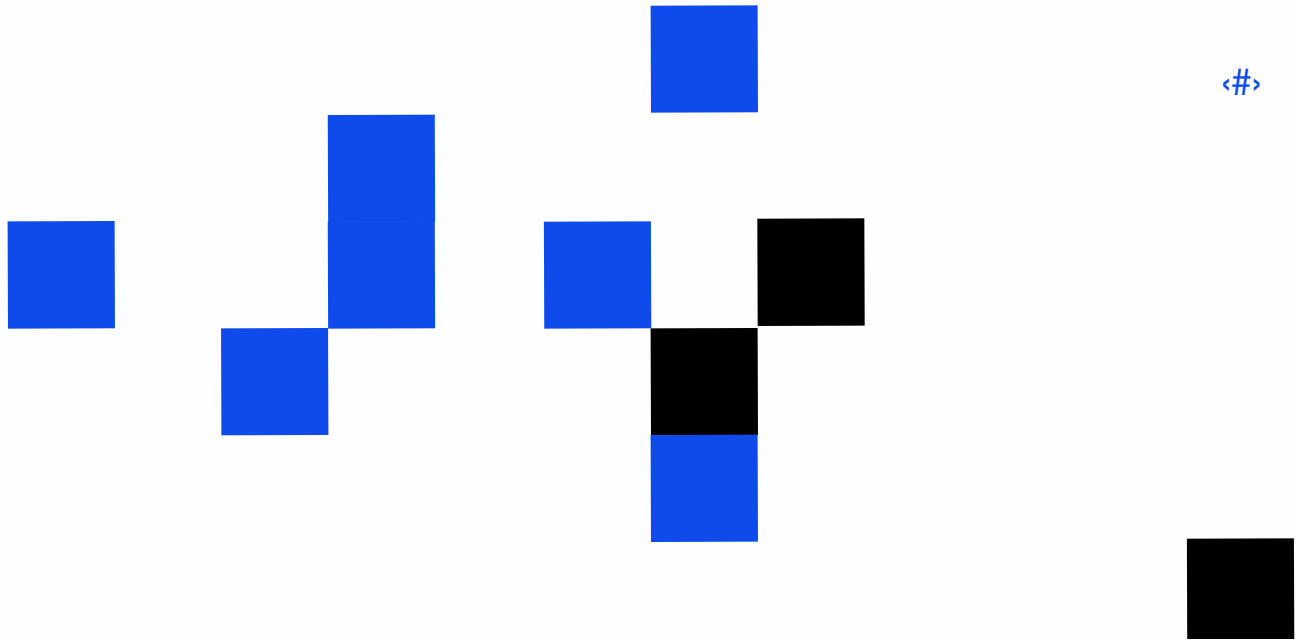
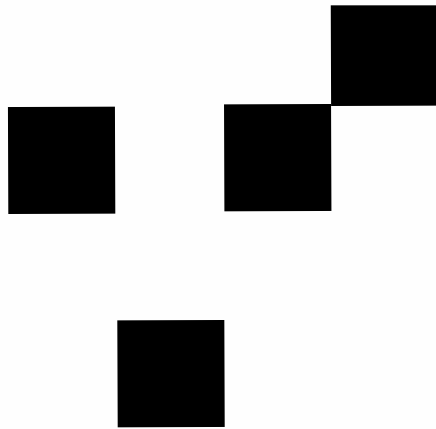
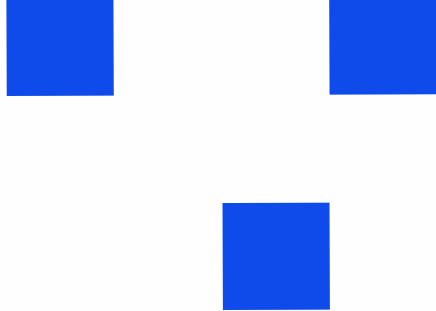
About Tutor (Tanzeem)

About Tutor (Zack)

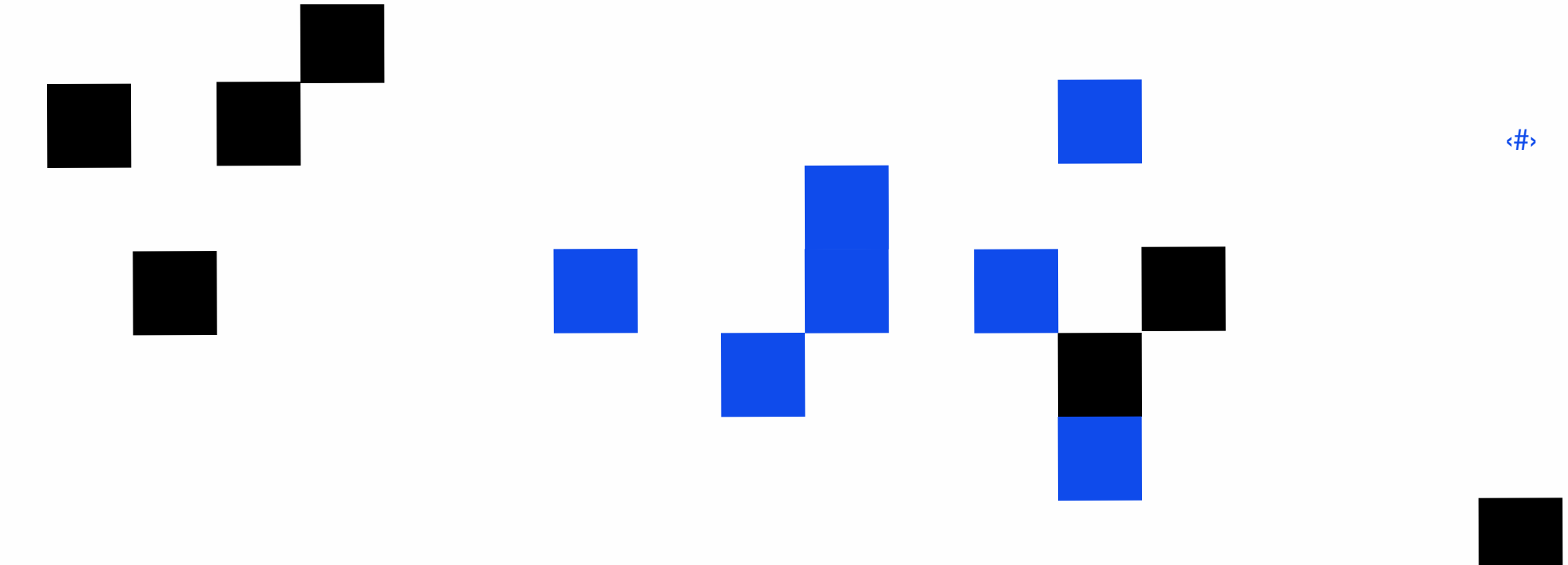
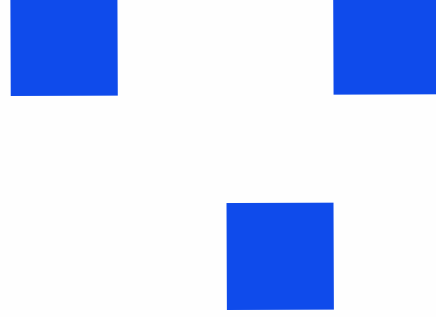
- Phd Candidate | BCI - Causal biomarkers for depression
- Interested in representation theory and computation.
- Still can't get my head around p vs np.
- Undergraduate qualifications;
 - Bachelor of computing science (major in AI and data science)
 - Diploma of Innovation (alternative to BCII)



About Tutor (Cyrus)



Software Innovation Studio



Software Innovation Studio

- • Software Engineering + Data Science + new tech => users' needs
 - ✓ Software Engineering is a **direct sub-field of engineering** and has an overlap with **computer science** and **management science**.
 - ✓ Identify and fit the **users' pain points/needs** via developed software
- What is studio?
 - ✓ You can LEVERAGE your **professional experience** and **skills**
 - ✓ You can learn by **doing what you do**
 - ✓ A place to gain maximum **collaborative and communication experience without worrying fail**
 - ✓ **Try to build your first own product in the market**

AI Ascent

Keynote

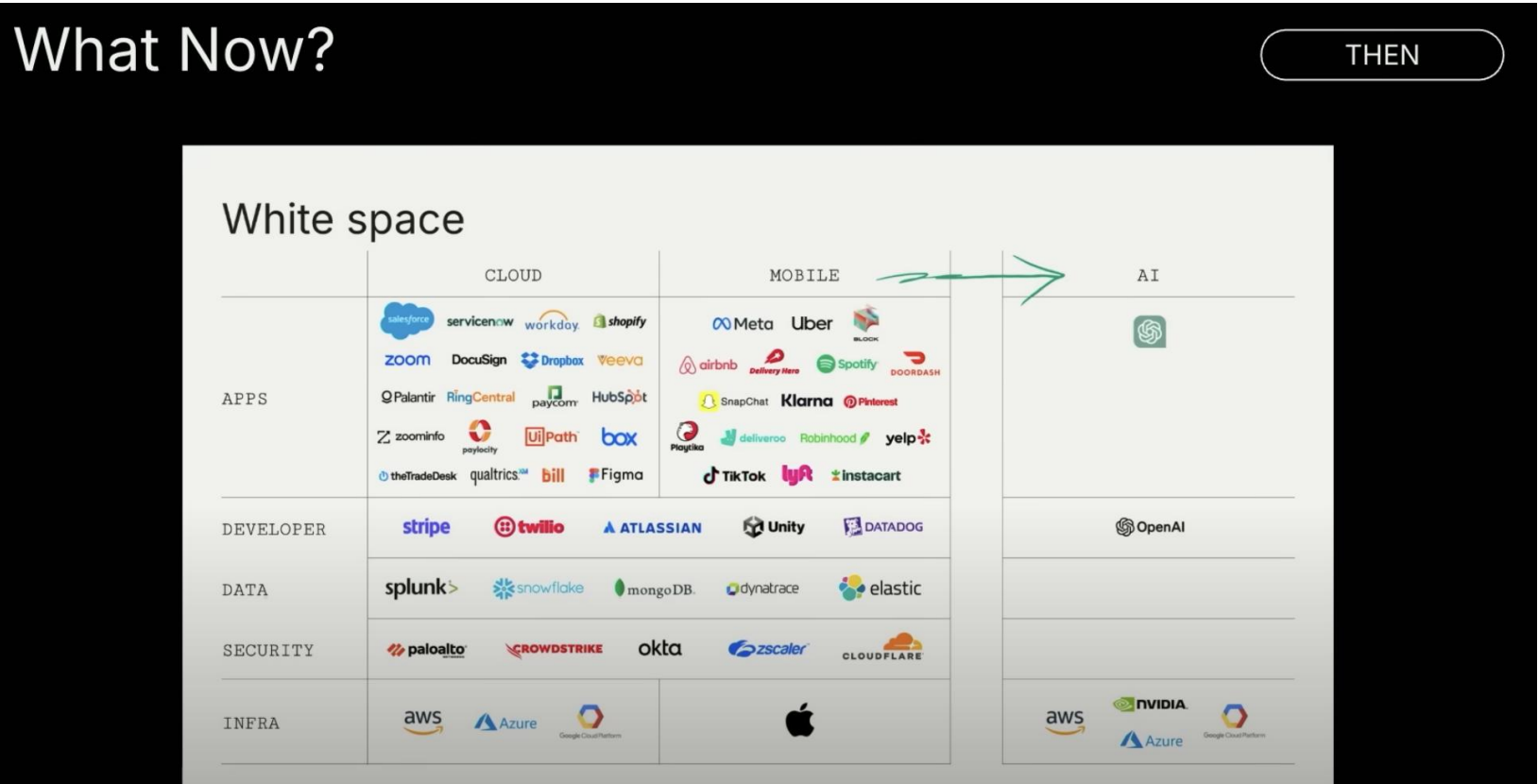
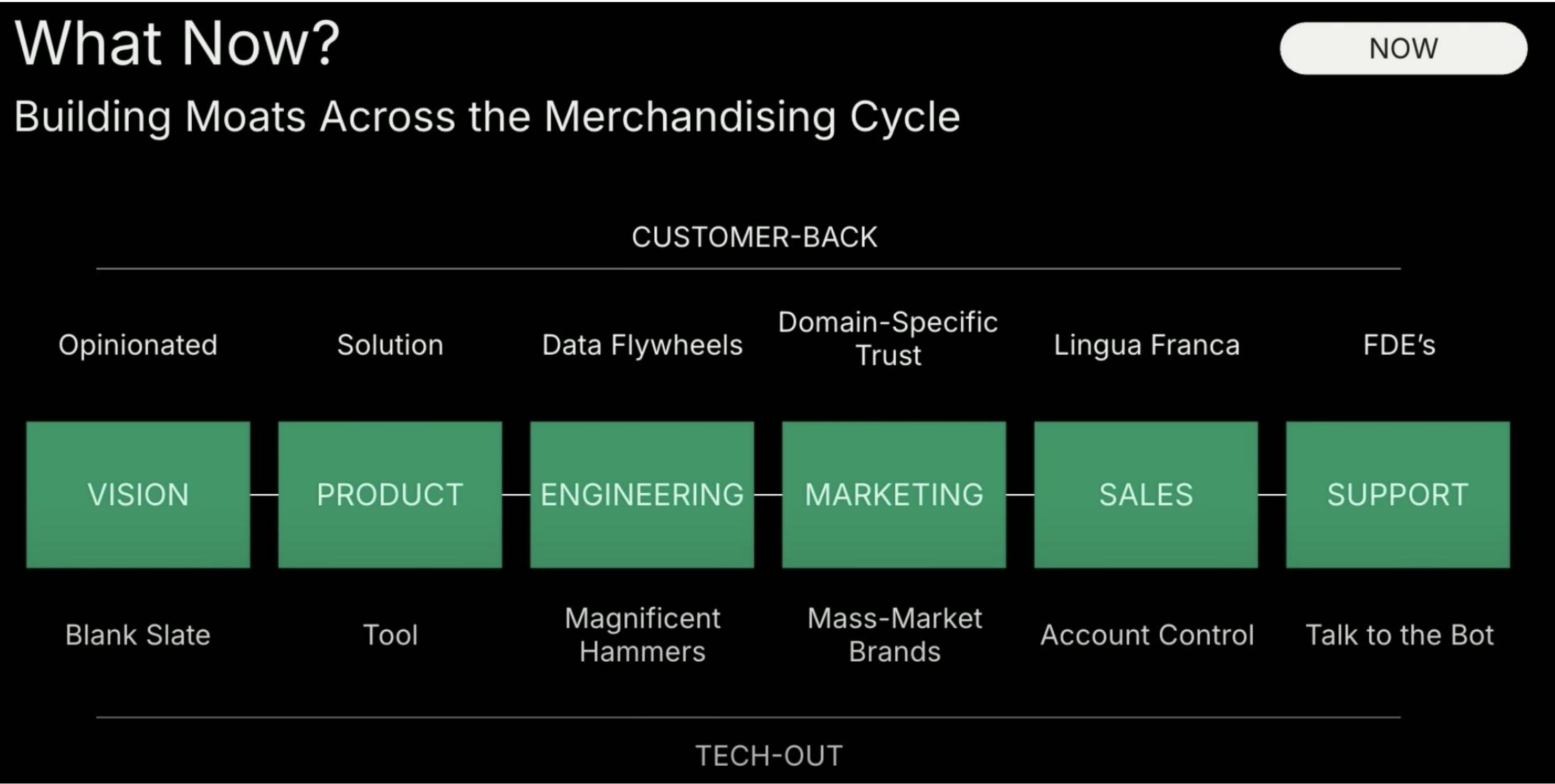
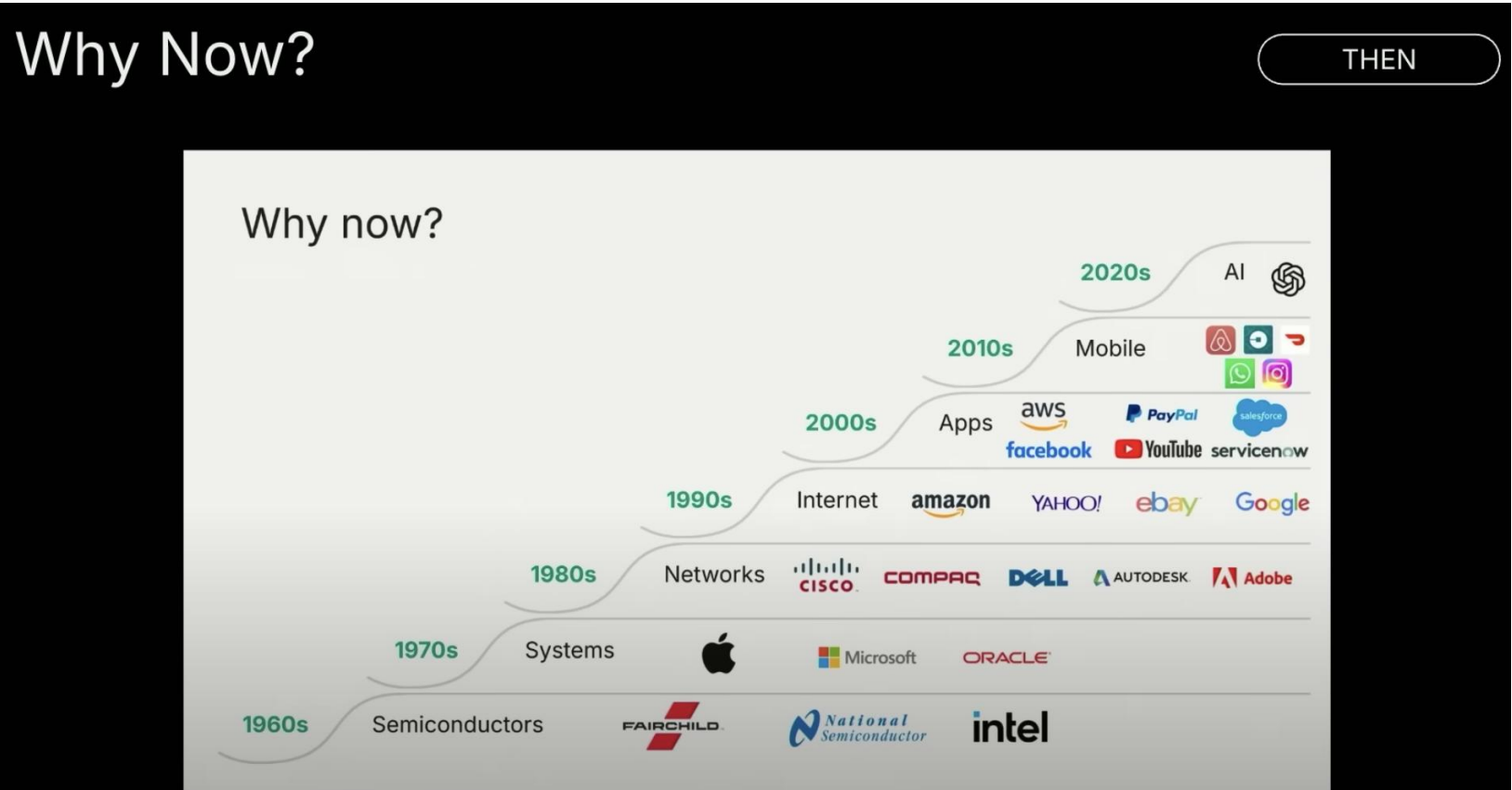
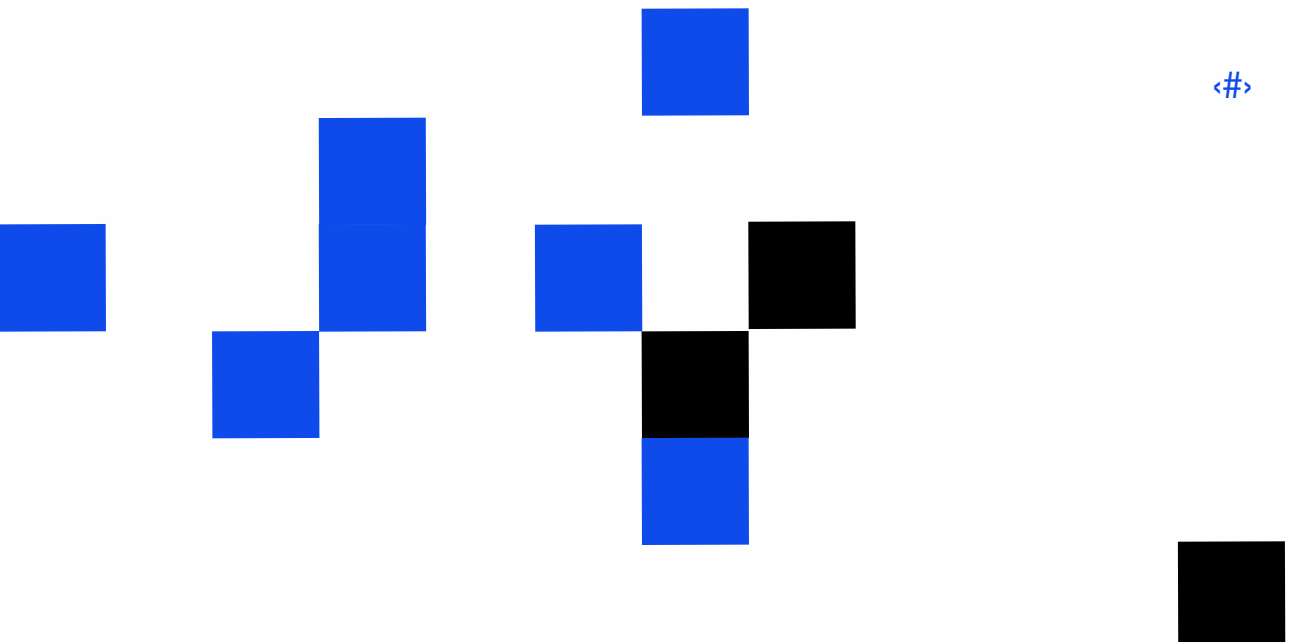
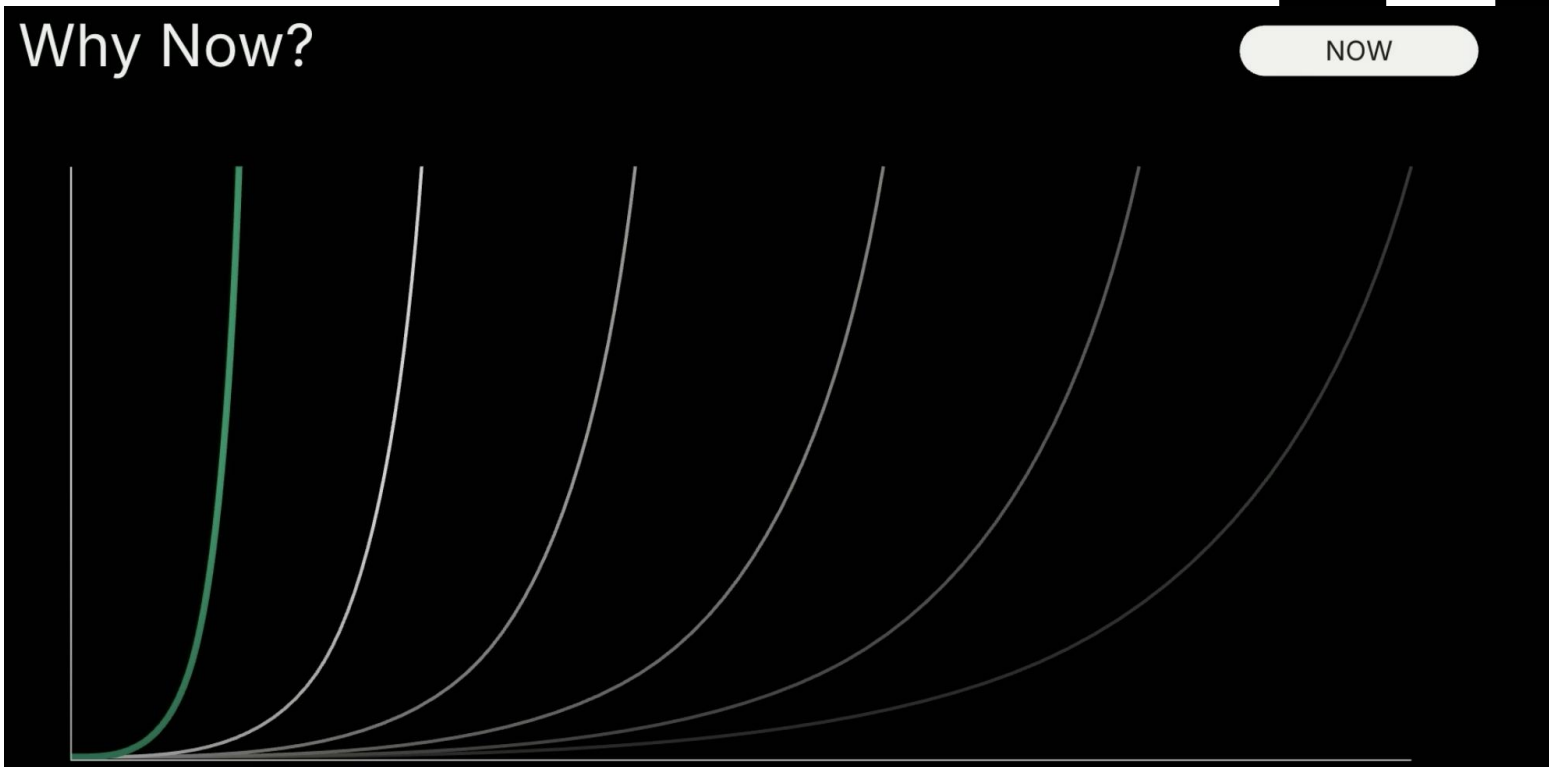
So What?

NOW

Services

AND

Software



https://www.youtube.com/watch?v=v9JBMnxuPX8&ab_channel=SequoiaCapital

The scopes



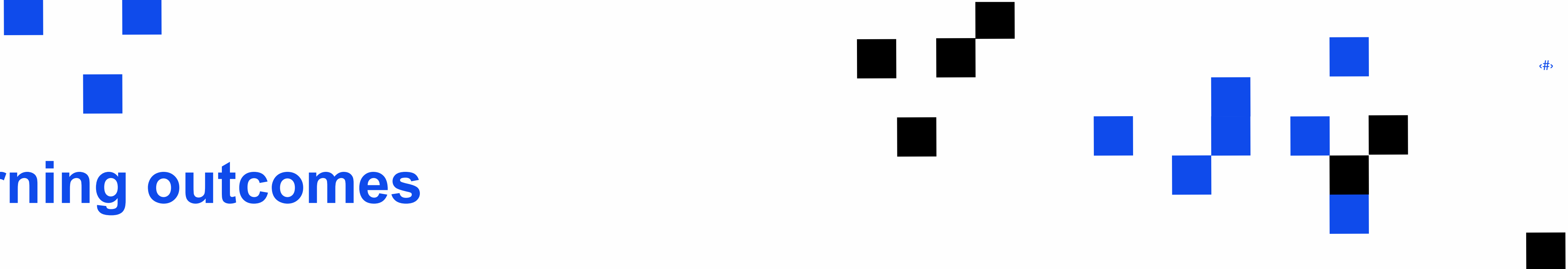
Idea(s) -> Product



Master Skills



Teamwork



Your learning outcomes

- • User-oriented project and its development
- Pitch the idea, full pitch deck
- Practical users' feedback (from classroom, Uni and/or market)
- Successful demo in week 12
- Roll out the product to market*
- Successful startup

* Bonus market in Assessment 4

Teaching and Learning Strategies

- • **Small** (~ 4-6 members) self-managed team:
 - + **Idea creation to fit users' needs perfectly (want to use it)**
 - + Programming and other skills practice
 - + Software engineering (**design thinking/analysis**, test, ...)
 - + Pitch, product pathways

Teaching and Learning Strategies

- Each team member is expected to have all followings each week

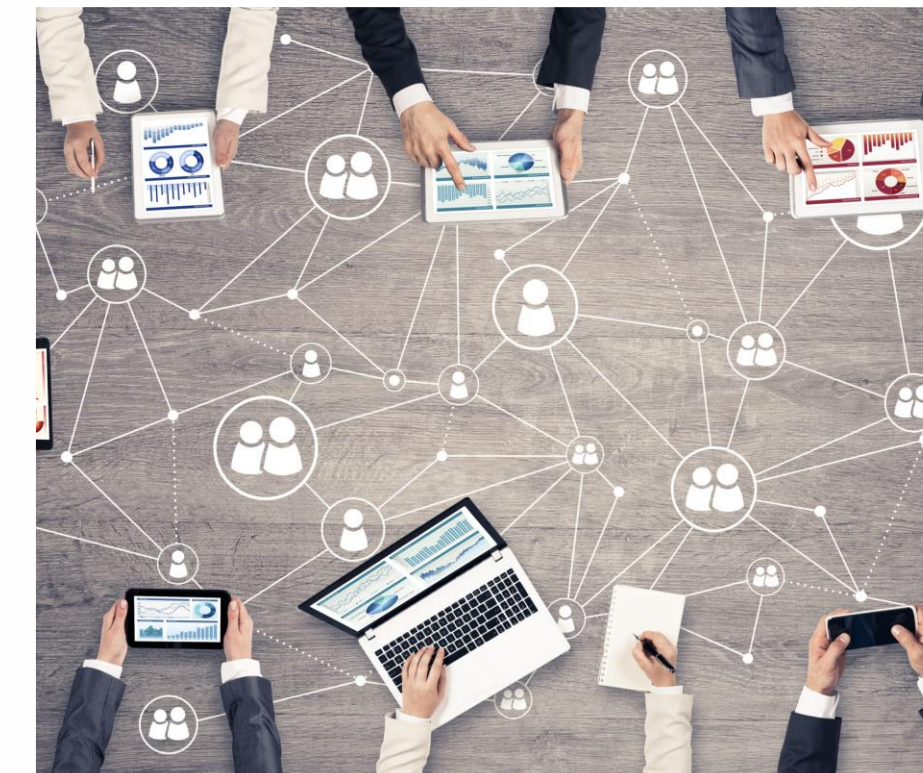


Weekly GitHub Update
Test Driven Development



Quality Weekly Meeting

Journal by 23:59 Thu (3-11)
Notes with clear actions



Clear Communications

Align progress clearly with
members and tutors weekly

Program in 2026 Spring

WJ: Weekly Journal (Individual, by end Thu)
MM: Meeting Minute (Group, by end Fri)
PM: Project Management (Group, by end Fri)

31 July	07 Aug	14 Aug	21 Aug	28 Aug	04 Sep	11 Sep	18 Sep	25 Sep	02 Oct	09 Oct	16 Oct	23 Oct	30 Oct
W 1	W 2	W 3	W 4	W 5	W 6	W 7	W8	StuVac	W 9	W 10	W 11	W 12	StuVac
Team	Team	Idea	Pitch	Vibe	Market	Market	Market	X	Cross	Market	Market	Demo	X
			WJ MM PM	WJ MM PM	WJ MM PM	WJ MM PM	WJ MM PM		WJ MM PM	WJ MM PM	WJ MM PM		

Sprint commence

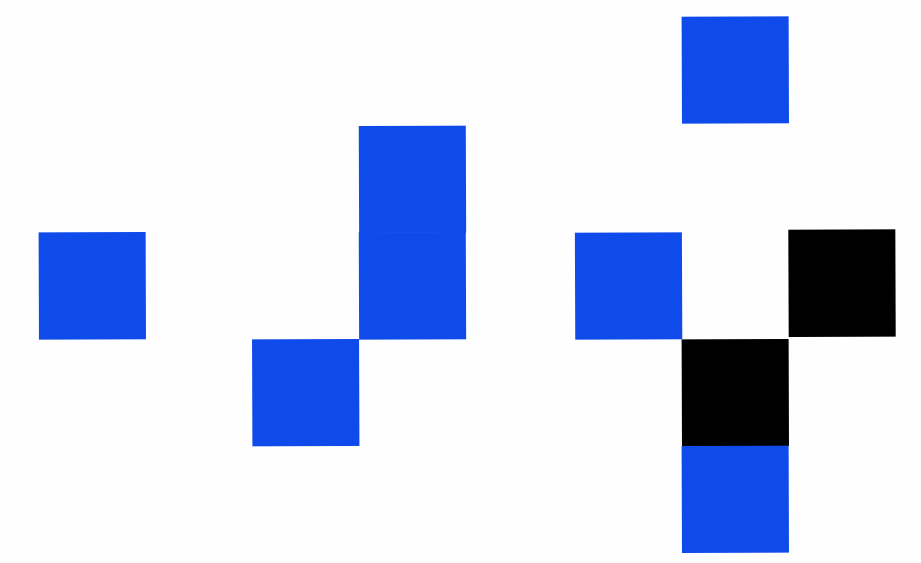
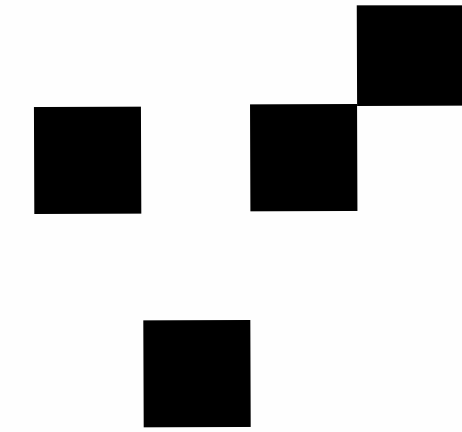
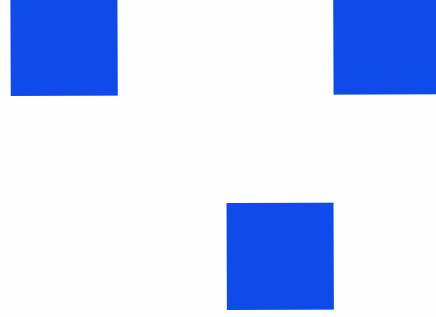
Assessment 1: Project Pitch
Slide 11:59 pm 20 Aug

Assessment 3-1: Peer Review
11:59 pm 17 Sep

Assessment 4: Slides and video
Assessment 5: Slides
11:59 pm 22 Oct

Assessment 2: 8 WJs + Design Thinking
Assessment 3-2: Peer Review
11:59pm 05 Nov

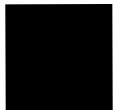
- ☐ Join workshop ontime
- ☐ Clear Teaming and Project Direction
- ☐ Engage team discussion actively
- ☐ Smooth teamwork and communications
- ☐ Engage MS Teams & Discussion Board
- ☐ Team-based decisions
- ☐ Finalise the tasks/jobs ontime
- ☐ Upload/circulate your work to mates ontime
- ☐ Get tech ready to the project and next journey
- ☐ Follow template to your work
- ☐ Project Pitch
- ☐ Market/competitor Analysis
- ☐ POC, Pitch Deck and Demo video
- ☐ User-oriented designs and implementations



<#>



Icebreaking and teaming



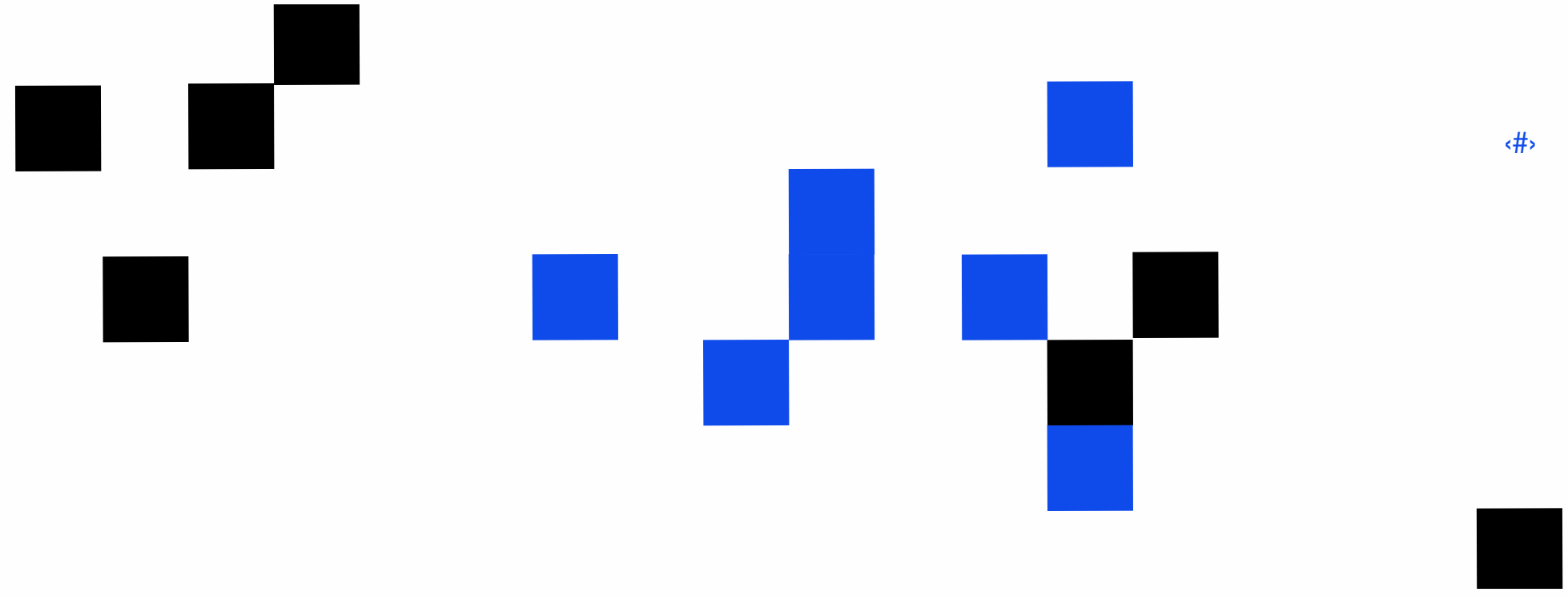
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Assessments

Assessments Overall

Assessment	Level	Tasks Group, Individual	Note
A1: Project Pitch	*	3-5 min pitch (10%)	Slides: 20 Aug Presentation: 21 Aug
A2: Weekly Journal + Design Thinking	*****	Weekly commitment and discussion Meeting preparation (10%) Comprehensive details (25%)	Week 4-11 Journal: end of Thu Design Thinking: 05 Nov
A3: Peer Review	***	Teamwork engagement (10%)	Peer Review: 17 Sep and 05 Nov
A4: Product Assessment	*****	1-2 min demo video and workable prototype (35%)	Group work, individual marking Video and slides: 22 Oct Demo in class: 23 Oct 2026
A5: Competitive Market Analysis	**	10 min presentation (10%)	Presentation in Weeks 6-11 Final slides: 22 Oct 2026



Assignments - Group

- • **Assignment 1** (**10%** group presentation, 3-5 min presentation)
 - + Team organisation, project planning (topic, scopes, timeline...), achievements, and team members' roles
 - + Slides due end of 20 Aug and have presentation in class
- **Assignment 5** (**10%** group presentation, ~10 min presentation)
 - + Target market
 - + Biggest competitor in the market and at least 2 killer function(s)
 - + Core values and Cons
 - + ...
 - + Slides due end of 22 Oct (final version)

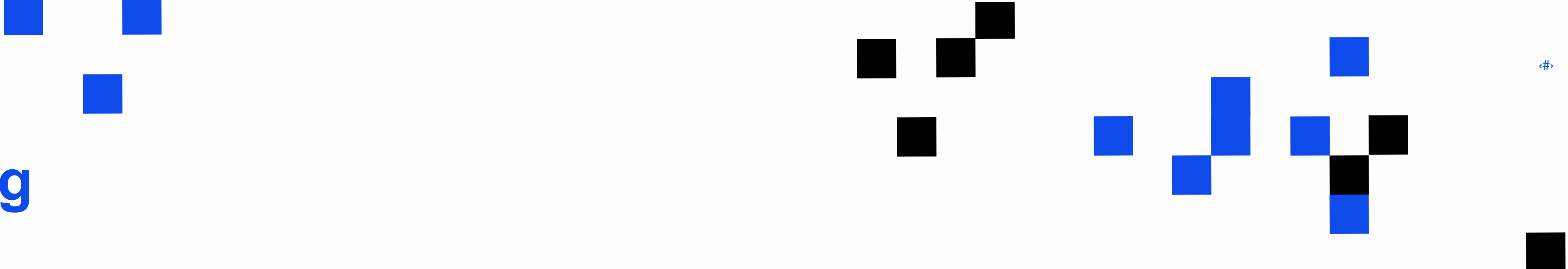


Assignments – Group with Individual Mark

- • **Assignment 4** (**35%** group presentation and demonstration)
 - + A successful showcase for your project
 - + Workable prototype demonstration with clear descriptions for topic, ideas, scopes, solutions, design, demonstrations, ...
 - + Draft slides and videos due end of 15 Oct
 - Slides and videos due by end of 22 Oct
 - Video and Prototype demo in class (23 Oct)

Assignments - Individual

- **Assignment 2** (**35%** individual, **1** final submission)
 - + Weekly Journal (10%) – Individual progress and meeting preparation
 - + Have **weekly** notes between weeks 4-11 (8 journals)
 - + First weekly journal will due by end of 20 Aug
 - + Design thinking for **1** your major development due by end of 5 Nov
- **Assignment 3** (**10%** individual, **2** peer reviews)
 - + Peer review about teamwork and engagement by SPARKPlus
 - + Your marks are only related to what you give (rate and comments)
 - + **Constructive advices/suggestions/comments are compulsory**
 - + End of 17 Sep and end of 5 Nov



Grading

- By lecturer and tutors

Weekly journals, Weekly meeting, project timeline, Trello, Git ...

Interactions with you or the team during workshops

Assess individual engagement and performance

Evaluate topic, scope, solution, design, achievements ...

- By peers

Review your team members's engagement and performance

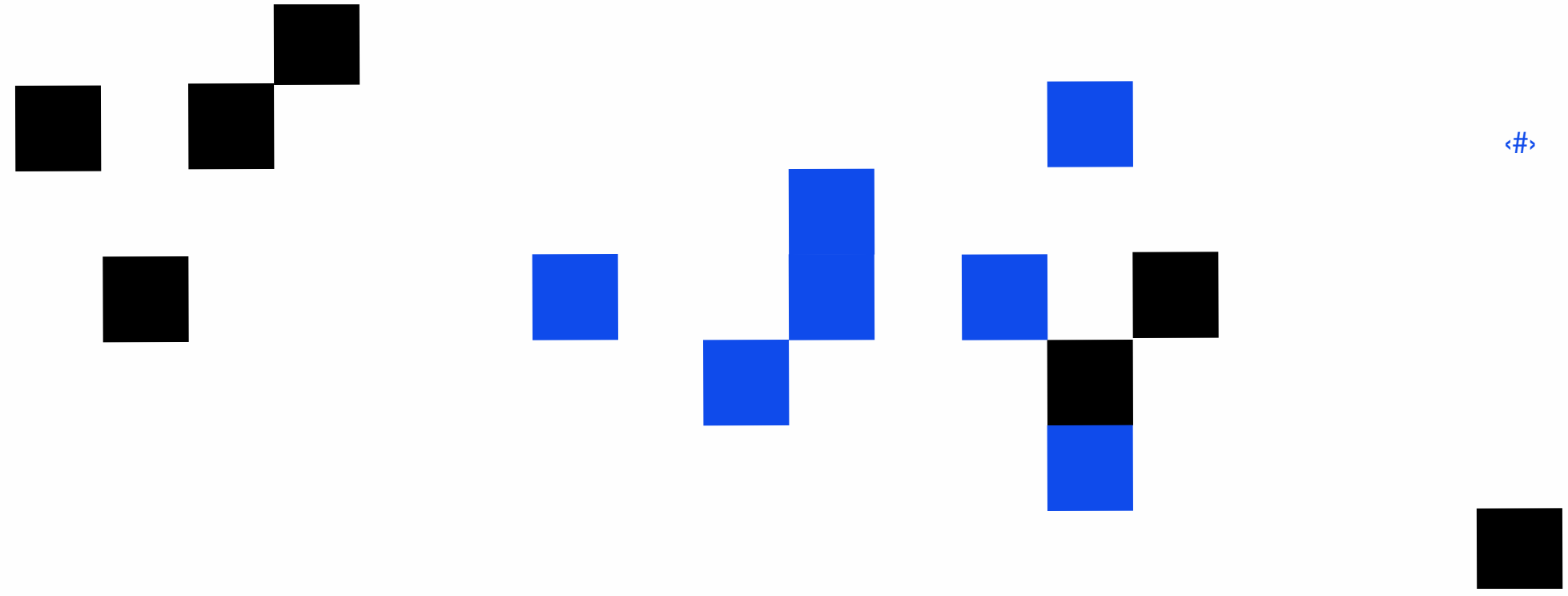
Share the rates and comments to individual teammates

Evaluate topic, scope, solution, design, achievements ...

Marking

- **Special consideration**
 - ✓ If your circumstances are eligible
 - ✓ Please apply it through online system with your full evidence **BEFORE** assessment deadline
- DO NOT simply email coordinator/tutor without results and evidence.

<https://www.uts.edu.au/current-students/managing-your-course/classes-and-assessment/special-circumstances/special-consideration>



If you have question

■ related to subject and assessments

- **Discussion Board** or **private channel** on MS Teams:
 1. General question -> Discussion Board
subject, assessment or other general issues
Any responses or interactions are welcome => **Extra points**
 2. Project related issues -> private channel
project, product, teamwork, management or team related issues
Any responses or interactions are welcome as well

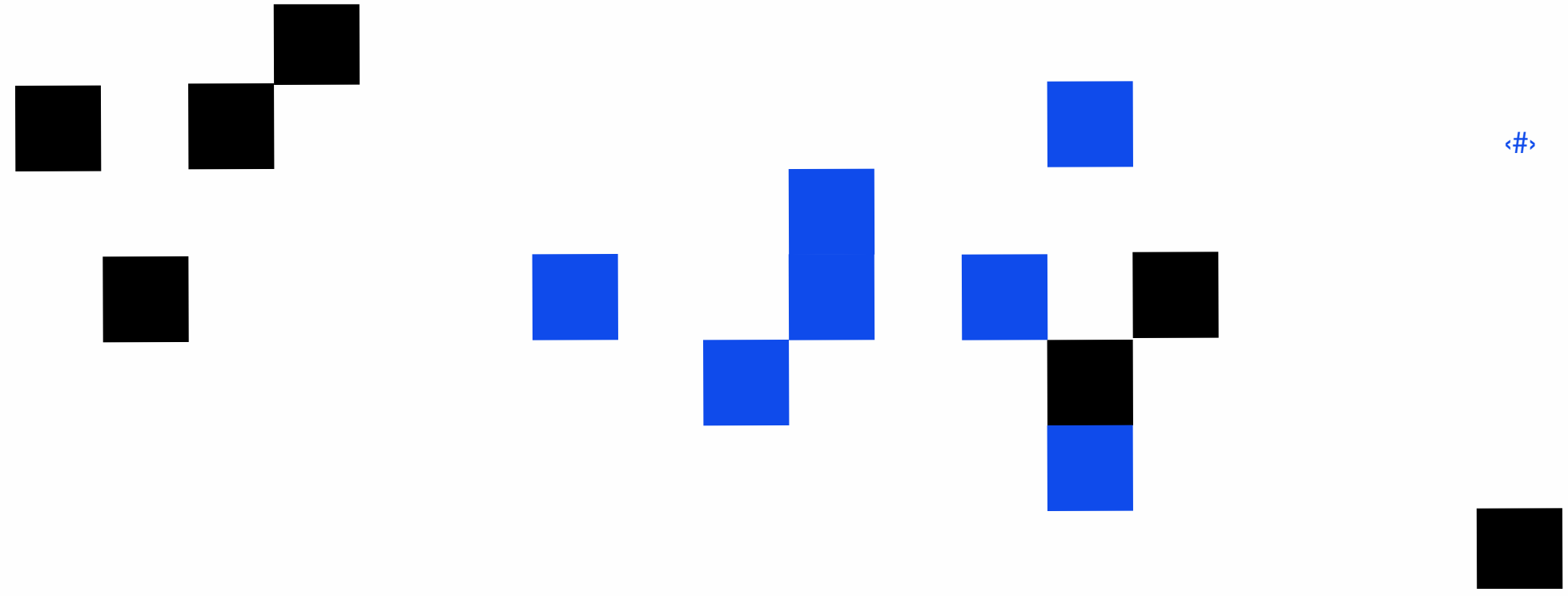


Communication with Tutor or Coordinator

- • **Email:**

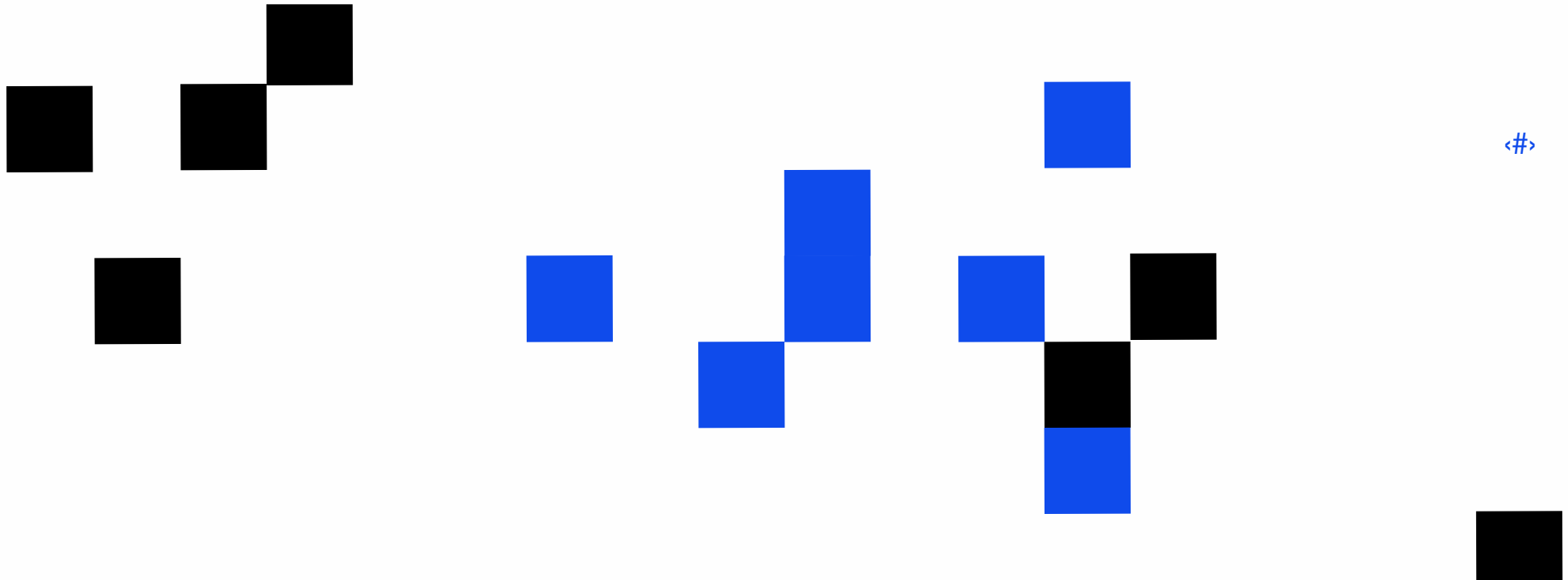
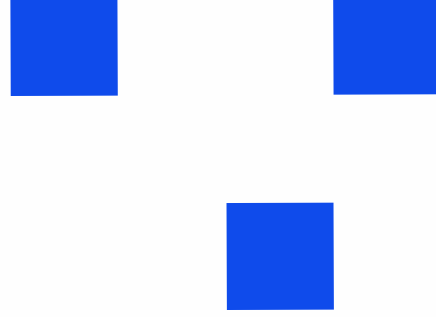
1. **ONLY** for personal stuff (illness, study/group problems, ...)
2. [41129] Subject of Email, your id

Example: [41129] **Get Sick 1234abcd**



Homework

- • Tutorials to Agile, Version Control, unit testing, Test Driven Development, Vibe Coding
- Background and field research for better design



Recommended reading materials

- • **Cracking the Coding Interview 189 Programming Questions and Solutions, 6th Edition**
- Any related videos, articles and/or research papers
- Software Engineering, 9th Edition, Sommerville, 2011
- Clean Code: A Handbook of Agile Software Craftsmanship, Robert C. Martin, 2008
- Clayton M. Christensen, Competing Against Luck: The Story Of Innovation And Customer Choice, 2016

Thank
you

